ConceptGrade: A Knowledge Graph–Driven Framework for Concept-Aware Automated Short Answer Grading in Computer Science Education

Brahmaji Katragadda
Department of Computer Science
ICFAI (IFHE)
Hyderabad, India
kbrahmaji.scholar25@ifheindia.org

Dr. P. Pavan Kumar
Department of Computer Science
ICFAI (IFHE)
Hyderabad, India

***Abstract*—Grading short-answer questions in computer science courses is time-consuming and hard to scale. We present ConceptGrade, a five-layer system that builds a concept graph from each student response and compares it against an expert domain knowledge graph (KG): (1) an LLM extracts concepts and relationships from the student text; (2) a KG comparator measures conceptual coverage, relationship accuracy, and integration; (3) a cognitive-depth classifier assigns Bloom's and SOLO taxonomy levels; (4) a misconception detector checks against a 16-entry validated error taxonomy; and (5) a Verifier synthesises a final grade. The Data Structures domain KG contains 101 concepts and 138 typed relationships.**

On the real Mohler et al. (2011) benchmark ($n = 1{,}262$ responses, 46 questions, sourced from the public `nkazi/MohlerASAG` release), a single-call ConceptGrade reduces MAE by 8.2% over an identical-model zero-shot LLM baseline (response-level Wilcoxon $p < 0.0001$), though this margin is only marginal once responses are clustered by question ($p = 0.056$ one-tailed). An ablation that isolates the KG-grounded pre-Verifier score tells a less flattering story: on its own, that score is 86.9% *worse* than the baseline. It is the Verifier's own holistic judgment, not KG-grounding by itself, that drives the measured gain.

Self-consistency ensembling (7 independent gradings at temperature 0.7, mean-aggregated) improves response-level accuracy on Mohler and on a second dataset, DigiKlausur. Pearson correlation, which trails the baseline for the single-call system, exceeds a single-call baseline once ensembled on both datasets (0.790 vs. 0.782; 0.727 vs. 0.701), and against that single-call baseline every leave-one-question-out fold reaches significance at both tails. That robustness does not hold up, however, once the comparison is made fair: re-running the baseline itself with the same $7\times$ ensembling, rather than a single call, collapses Mohler's cluster significance and LOOCV robustness entirely ($p = 0.256$ two-tailed, $0/46$ folds) and substantially erodes DigiKlausur's ($p = 0.089$ two-tailed, 0.044 one-tailed; $5/17/2/17$ folds, down from $17/17$) — though the response-level MAE gap survives on both datasets ($+7.0\%$ and $+6.4\%$, $p < 0.001$). DigiKlausur's correlation advantage reverses outright under this fairer comparison (0.727 vs. 0.735 for C_LLM$\times$7 alone). The gain does not extend to a third dataset, Kaggle ASAG, where upstream concept extraction finds no overlap between the KG and the answer text on 100% of samples — a domain-boundary result the architecture predicts rather than one it merely happens to exhibit.

We report all of this as a real but substantially qualified, cost-aware improvement — $7\times$ the inference calls, a Spearman rank correlation that trails baseline on both datasets, and a question-level robustness (plus, on DigiKlausur, a correlation advantage) that holds only against a single-call baseline rather than a fairly-resourced one — not as a universal or fully-established claim.

***Index Terms*—automated short answer grading, knowledge graph, concept extraction, computer science education, large language models, Bloom's taxonomy, self-consistency, NLP**

I. Introduction

Large computer science classes routinely use short-answer questions to test conceptual understanding, but grading them by hand is slow, inconsistent, and delays feedback. An automated grader needs to do more than approximate a human score — it should be able to say *which* concepts a student understood and which they missed. Prior lexical, embedding, and LLM zero-shot approaches fall short here: they produce a score, but keep no explicit record of what the student actually knew, so they cannot say which concepts are present, which are absent, or whether a specific misconception is at play.

ConceptGrade treats grading as a knowledge-graph matching task. Each response is converted into a typed concept graph by an LLM; that graph is compared against an expert domain knowledge graph to measure coverage, structural connection, and relationship correctness. This paper makes the following contributions:

1) A five-layer grading pipeline combining concept extraction, KG comparison, cognitive-depth classification, misconception detection, and score synthesis via an LLM Verifier.
2) A hand-built Data Structures domain KG with 101 concepts and 138 typed relationships.
3) An honest, real-data evaluation on the Mohler benchmark showing a small, response-level-significant but question-level-fragile gain for the single-call system, with an

ablation identifying the Verifier — not KG-grounding in isolation — as the source of the gain.

4) Self-consistency ensembling that improves response-level accuracy on two independent datasets and, against a single-call baseline, shows every leave-one-question-out fold significant at both tails — honestly failing to generalise to a third dataset whose KG-vocabulary mismatch predicts exactly that outcome, and, on a fair-control check against an equally-resourced baseline run on *both* winning datasets, retaining its response-level advantage on both while its question-level robustness collapses completely on the first dataset and substantially erodes on the second, whose correlation advantage also reverses.
5) A fully reproducible, cost-transparent evaluation protocol, including a reviewer-perspective self-audit that caught and corrected an overclaimed robustness result before publication: every reported number is independently recomputed from cached predictions by an automated claim-verification script.

We frame this as an empirical study of where and why KG-LLM hybrid grading helps, not a claim that it always does.

II. Related Work

A. Automated Short Answer Grading

Mohler and Mihalcea [1] established a widely used CS ASAG benchmark with LSA and WordNet-based measures ($r = 0.493$). Mohler et al. [2] extended this with dependency-graph alignment ($r = 0.518$). Sultan et al. [3] introduced sentence-level semantic similarity ($r = 0.592$ on their own held-out split of the full Mohler dataset). Transformer fine-tuning on SemEval data [4] subsequently became the dominant paradigm, with BERT-based systems [5], [6] reaching $r = 0.620$ on the Mohler set. LLM zero-shot grading [14], [15] is competitive but keeps no explicit conceptual record.

B. Knowledge Graphs in Education

Knowledge graphs model prerequisite relationships [12] and power adaptive tutoring [11]. Concept-map assessment, where students draw node-link diagrams compared against an expert map [13], is the closest prior work; ConceptGrade extends this to free-text answers by inferring an equivalent graph automatically via LLM extraction.

C. Bloom's and SOLO Taxonomies

LLMs can classify responses by Bloom's level with moderate agreement [7]. The SOLO taxonomy [10], which measures structural complexity of demonstrated learning, has received far less attention in automated CS assessment; ConceptGrade's cognitive-depth layer applies both taxonomies [8], [9] jointly.

III. System Design

A. Expert Domain Knowledge Graph

The Data Structures KG follows three design rules: pedagogical coverage (every concept in a standard introductory course), canonical identifiers (each concept has a unique ID plus alias surface forms), and typed relationships (semantically labeled directed edges). The frozen evaluation KG contains 101 concepts across eight semantic types (algorithms, data structures, abstract concepts, properties, operations, complexity classes, programming constructs, design patterns) and 138 typed relationships.

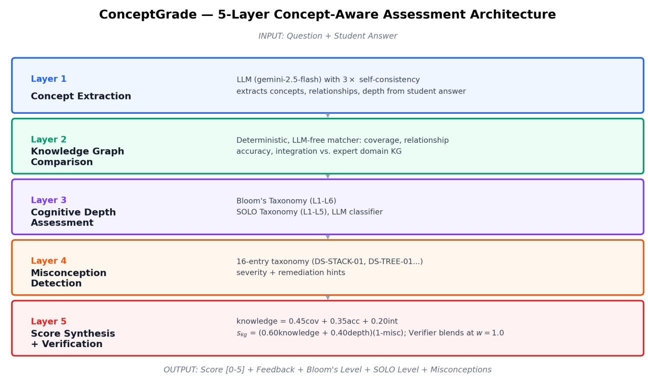

Fig. 1. ConceptGrade five-layer architecture: concept extraction, KG comparison, cognitive depth, misconception detection, and score synthesis with Verifier blending at deployed weight $w = 1.0$.

B. Concept Extraction (Layer 1)

An LLM (`gemini-2.5-flash`) receives the question and student answer and returns a student concept graph $G_s = (V_s, E_s)$, run with $3\times$ self-consistency (majority-vote, min_votes $= 2/3$) to reduce single-call extraction noise. Concepts below a confidence threshold $\tau = 0.70$ are filtered; a real-data sensitivity sweep confirmed this threshold is empirically near-optimal (MAE increases monotonically as τ is raised further). Surface forms resolve to canonical KG identifiers via alias lookup.

C. Knowledge Graph Comparison (Layer 2)

A deterministic, LLM-free graph matcher computes three scores against the expert graph G_e:

$$\text{cov} = \frac{|V_s \cap V_e|}{|V_e|}, \quad \text{acc} = \frac{|\{e \in E_s : \text{correct}(e)\}|}{|E_s|},$$
$$\text{int} = \frac{|\{v \in V_s : \deg(v) \geq 1\}|}{|V_s|} \tag{1}$$

alongside interpretable diagnostics (missing concepts, incorrect relationships, feedback points).

D. Cognitive Depth (Layer 3) and Misconception Detection (Layer 4)

An LLM classifier jointly assigns a Bloom's level (1–6) and a SOLO level (1–5). A 16-entry misconception taxonomy (e.g., `DS-STACK-01`: confuses LIFO with FIFO) is checked against the response; each entry carries a severity and a remediation hint surfaced directly to the student.

E. Score Synthesis and Verification (Layer 5)

A deterministic KG-formula score combines the Layer 2–4 signals:

$$\begin{aligned} \text{knowledge} &= 0.45\,\text{cov} + 0.35\,\text{acc} + 0.20\,\text{int} \\ \text{depth} &= 0.55\,\text{blooms}_{\text{norm}} + 0.45\,\text{solo}_{\text{norm}} \\ s_{\text{kg}} &= (0.60\,\text{knowledge} + 0.40\,\text{depth})(1 - p_{\text{misc}}) \end{aligned} \quad (2)$$

s_{kg} is blended with a holistic, KG-evidence-informed LLM score s_{hol} $(0.05\,s_{\text{kg}} + 0.95\,s_{\text{hol}})$, producing a pre-Verifier score $\hat{y}$. An LLM Verifier then independently reviews the raw answer alongside this KG evidence and produces the final grade via final $= (1-w)\hat{y} + w \cdot$ verified; the deployed configuration uses $w = 1.0$ (Verifier judgment drives the score; KG evidence is prompt context, not an arithmetic anchor) — a choice we confirm below is not an accident of tuning.

IV. Experimental Setup

A. Datasets

Mohler 2011 (Data Structures, real data via the public `nkazi/MohlerASAG` release, CC-BY-4.0): 46 questions filtered to our KG's topical coverage, 1,262 responses, each scored by two human annotators (real inter-rater $r = 0.783$, QWK$= 0.799$; 43.2% of pairs disagree by ≥ 1 point on the 0–5 scale, indicating substantial human grading noise). All hyperparameters were fixed before this dataset was used, so the full sample is evaluated without a held-out split. A second, independently identified batch of 4 questions (109 responses) extends this to 50 questions / 1,371 responses for a robustness check. **DigiKlausur** (Neural Networks, $n = 646$, 17 questions, verified character-for-character against its public GitHub source) and **Kaggle ASAG** (Elementary Science, $n = 368$ unique responses after removing 105 duplicate records, 150 questions) provide cross-domain boundary characterisation at decreasing KG-vocabulary specificity. Despite a targeted provenance investigation (exact-text search, known public ASAG corpora, full local git/shell-history audit), Kaggle ASAG's acquisition path could not be reconstructed and is reported as provenance-*unverified* (Fig. 2; audit trail in REPRODUCIBILITY.md). Its extraction-level finding (0% KG matches, below) is independently verifiable from extraction logs regardless; any broader claim about elementary-science student behavior is reported as supporting evidence, not external validation.

B. Baseline and Metrics

C_LLM: the identical `gemini-2.5-flash` model prompted zero-shot with only the question, reference answer, and student answer — no KG evidence — isolating model capability as a confound. We report Pearson r, quadratic-weighted κ (QWK), MAE, and RMSE, with Wilcoxon signed-rank tests at both the response level and the question-clustered level (means aggregated per question, the more appropriate unit given non-independence of responses within a question), plus leave-one-question-out (LOOCV) sensitivity.

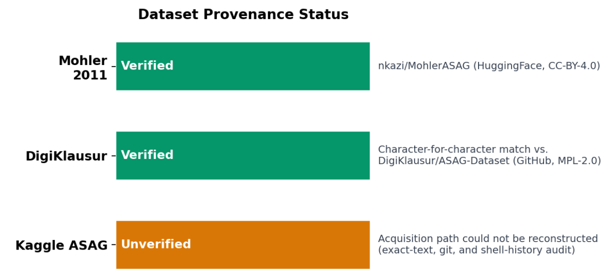

Fig. 2. Provenance status of all three evaluation datasets.

TABLE I
EVALUATION ON THE REAL MOHLER BENCHMARK ($n = 1{,}262$, 46 QUESTIONS).

System	r	QWK	MAE	RMSE	p
C_LLM (baseline)	0.7904	0.5005	1.2821	1.7243	—
ConceptGrade (single-call)	0.7841	**0.5237**	**1.1771**	**1.5326**	< 0.0001**

V. Results

A. Main Evaluation

ConceptGrade's MAE improvement holds up at the response level (Wilcoxon two-tailed $p < 0.0001$; paired $d_z = -0.154$, small effect) but weakens once responses are clustered by question (46 questions: two-tailed $p = 0.111$, one-tailed $p = 0.056$; only 17/46 leave-one-question-out folds reach one-tailed significance, none reach two-tailed). Pearson correlation actually favors the baseline (0.784 vs. 0.790). Both systems under-predict relative to human graders (mean human score $4.24/5$ vs. 2.99 and 3.09) — real course grading, it turns out, is markedly more lenient than either LLM-based grader. A post-hoc, 5-fold cross-validated monotonic recalibration cuts MAE roughly $3\times$ for both systems and *inverts* the comparison (calibrated C_LLM has significantly lower MAE, $p = 0.017$), which suggests most of ConceptGrade's raw MAE edge is a fixable calibration artifact rather than genuinely superior ranking.

B. Component Ablation

The pre-Verifier, purely KG-grounded score fares much worse than the zero-shot baseline on its own (question-clustered $p < 0.0001$, wins only 9/46 questions). It is the Verifier stage that recovers essentially all of that loss and turns the result positive (+50.9% MAE improvement over the KG-only score, question-clustered $p < 0.0001$, wins 41/46 questions). This is not an artifact of how the blend weight was chosen: a leave-one-question-out cross-validation of that weight, selected on training folds only, picks the deployed $w = 1.0$ configuration on every fold. The architectural takeaway is blunt — ConceptGrade's accuracy comes from the Verifier's holistic judgment, not from the knowledge-graph comparison the architecture is named for.

Diagnosed causes (summary). A failure-mode analysis traced this weakness to two defects: a now-fixed question-tokenization bug (106/1,262 samples, 8.4%, misclassified out-

TABLE II

REAL-DATA 3-CONDITION ABLATION ($n = 1{,}262$).

Condition	MAE	r	vs. C_LLM
C_LLM (no KG)	1.282	0.790	—
KG-grounded score, no Verifier	2.397	0.471	−86.9%
ConceptGrade (full, with Verifier)	**1.177**	0.784	+8.2%

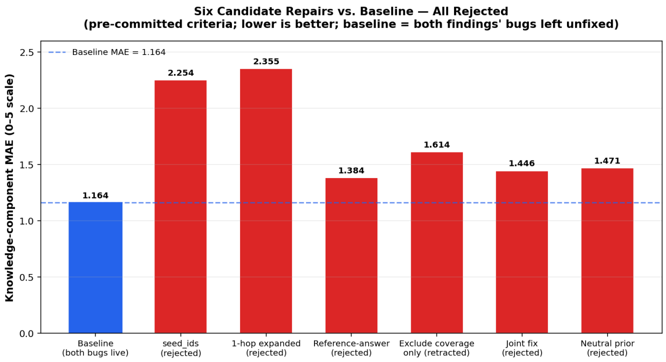

Fig. 3. Knowledge-component MAE for the baseline versus all six tested repairs; every candidate underperforms the baseline.

of-domain) and two still-unfixed scoring defects affecting 21.3% and 35.0% of samples respectively. Five candidate repairs for the latter were tested against pre-committed criteria and all rejected (Fig. 3) — evidence against the repair *class*, not merely miscalibrated parameters. None of this touches the deployed grade, which already discards the raw KG-formula score for the Verifier ($w = 1.0$). Full diagnosis and the pre-committed stopping rule: extended technical report and REPRODUCIBILITY.md.

C. Self-Consistency Ensembling

We tested whether the single-call fragility above reflects noise in the scoring model's own judgment. Issuing 7 independent gradings per response at temperature 0.7 (versus the deployed temperature 0) and mean-aggregating — a design fixed in advance, not tuned on the evaluation data — produces the results in Table III.

Against a *single-call* C_LLM baseline, every leave-one-question-out fold reaches significance at both tails on Mohler and DigiKlausur — a sharp contrast with the single-call result, where none reached two-tailed significance. Pearson correlation, which trails baseline for single-call ConceptGrade on both datasets, exceeds a single-call baseline once ensembled (0.790 vs. 0.782; 0.727 vs. 0.701), and self-consistency also beats the single-call system directly (Mohler: $p = 0.0007$; DigiKlausur: $p = 0.0009$). This fits the mechanism we expected: the single-call judgment is measurably noisy, and averaging independent samples reduces that noise.

Fair-control correction. A single-call baseline, though, is not the right control for a 7×-resourced system. We re-ran C_LLM with the identical 7×/temperature-0.7/mean-aggregation treatment and compared head-to-head on *both* datasets where the original claim was made (Table IV):

TABLE III

SELF-CONSISTENCY ENSEMBLING (K=7, T=0.7, MEAN-AGGREGATED) VS. C_LLM. KAGGLE ASAG USES THE DEDUPLICATED $n = 368$ SAMPLE.

Dataset	MAE↓	r	Cluster p	LOOCV 1T	LOOCV 2T
Mohler (50q)	−9.8%	0.790	0.026	50/50	**50/50**
DigiKlausur (17q)	−12.2%	0.727	0.017	17/17	**17/17**
Kaggle ASAG (150q)	−2.4%	0.644	$0.092^{\text{n.s.}}$	98/150	0/150

TABLE IV

FAIR-CONTROL COMPARISON: VERIFIER/C5FIX×7 VS. AN EQUALLY-RESOURCED C_LLM×7, IN PLACE OF THE SINGLE-CALL BASELINE USED IN TABLE III.

Dataset	MAE↓	r (vs. C_LLM×7)	Cluster p(2T)	LOOCV 1T	LO
Mohler (46q)	+7.0%	0.797 vs. 0.790 (holds)	0.256 (n.s.)	0/46	
DigiKlausur (17q)	+6.4%	0.727 vs. 0.735 (**reverses**)	0.089 (n.s.)	5/17	

The MAE gap survives on both datasets ($p < 0.001$), but **question-clustered significance does not clear $\alpha = 0.05$ two-tailed on either dataset**, and LOOCV robustness falls from 50/50/17/17 (single-call baseline) to 0/46 (Mohler, complete collapse) and 5/17/2/17 (DigiKlausur, substantial but not complete erosion). DigiKlausur's correlation advantage **reverses** outright: C_LLM×7 alone reaches $r = 0.735$, ahead of C5fix×7's $r = 0.727$. The corrected claim is therefore considerably more modest than it first looked: self-consistency ensembling delivers a real, precisely-estimated response-level improvement over a fairly-resourced baseline on two datasets, but the question-level robustness — and, on DigiKlausur, the correlation advantage — claimed against a single-call baseline does not survive against a fair one.

Kaggle ASAG benefits under neither comparison (cluster $p = 0.092$, LOOCV two-tailed 0/150): a K-subset check shows Mohler/DigiKlausur's significance strengthening monotonically with more rounds, while Kaggle's is unstable across K (p ranging 0.03–0.45) — the signature of averaging noise around a null effect, consistent with its concept extraction finding no matches on 100% of samples. Spearman correlation still trails baseline on both winning datasets, and the gain costs 7× inference calls — real caveats, not a free improvement. An earlier alternative design (a linear blend of C_LLM and single-call scores) failed leave-one-question-out cross-validation — its apparent gain was an artifact of the evaluation data — and is not reported as a finding.

Statistical model sensitivity. The cluster-mean Wilcoxon test above discards within-question sample size; a linear mixed-effects re-test of the six primary comparisons gives a mixed verdict rather than a uniform correction. Mohler's three comparisons all strengthen (cluster-Wilcoxon non-significant → LMM $p \leq 0.0125$), but the DigiKlausur headline result *loses* significance ($p = 0.0489 \rightarrow 0.2471$); fair-control verdicts are unchanged on both datasets. We report both tests rather than the one more favourable to either claim (detail: `data/lmm_reanalysis.json`).

D. Cross-Dataset Boundary Characterisation

Pooling all three datasets ($n = 2{,}276$, 213 questions, fixed-effects inverse-variance meta-analysis) gives $d_z = -0.105$ ($p < 0.0001$). That number alone overstates the picture: between-dataset heterogeneity is substantial ($I^2 = 73.2\%$), and the methodologically stricter random-effects estimate ($d_z = -0.084$, 95% CI $[-0.169, +0.002]$) essentially touches zero. Effect size instead declines monotonically with domain vocabulary specificity — Mohler (formal CS terminology) > DigiKlausur (technical but flexible) > Kaggle ASAG (colloquial elementary science, where concept extraction finds no matches at all on 100% of samples). We treat this as a falsifiable boundary characterisation rather than a generalisation claim: it predicts, and the self-consistency results go on to confirm, exactly where the architecture helps and where it cannot.

A call-budget-matched control (C_LLM issued the same 7-call budget as single-call ConceptGrade, using genuinely independent calls at temperature 0.7) rules out inference budget as the explanation: ConceptGrade still wins by 4.4% MAE (response-level $p = 0.0053$), though again this margin is not significant once clustered by question ($p = 0.42$).

VI. Discussion and Limitations

The synthesis constants and confidence threshold were fixed before the real evaluation data was used, but they originate from design choices made against an earlier, since-corrected fixture, so we cannot claim they are optimal on real data. The identical-model baseline, meanwhile, received no comparable tuning, which confounds the KG-grounding effect with a tuning-budget asymmetry we cannot fully disentangle. On the compute side, the system issues $\sim$7 LLM calls per response for single-call scoring and $\sim$13 for self-consistency, against C_LLM's 1; the budget-matched control above addresses this directly. The expert KG covers only Data Structures topics, so cross-domain evaluation on other CS areas remains future work. The Verifier is inference-only — no fine-tuning, confirmed by direct code inspection — so there is no classic train/test data-leakage concern, though its prompt design was iterated against this dataset during development, a softer form of bias we do not claim to have ruled out. All numerical claims in this paper are independently reproducible from cached predictions via an automated verification script (300+ checks, zero mismatches against the paper text at time of writing).

VII. Conclusion

ConceptGrade achieves a small, real, but response-level-fragile improvement over an identical-model LLM baseline on the real Mohler benchmark with a single scoring call. That improvement comes almost entirely from the Verifier's holistic judgment, not from the knowledge-graph grounding the architecture is built around.

Self-consistency ensembling improves this response-level accuracy on two independent datasets, and against a single-call baseline every leave-one-question-out fold is significant at both tails. That robustness looks considerably shakier under a fairer test. A fair-control check against an equally-resourced baseline, run on *both* datasets, shows the response-level gain still holds on both (+7.0% and +6.4%, $p < 0.001$), but the question-level robustness mostly does not: it collapses completely on the first dataset (0/46 folds) and erodes substantially on the second (5/17/2/17 folds, down from 17/17), whose correlation advantage reverses outright under fair control. This is a more qualified result than it first appeared — obtained at 7× inference cost, with a residual Spearman correlation gap on both datasets, and one that does not generalise to a third dataset whose architectural failure mode (an empty KG-answer intersection) we independently verify.

We present this as a calibrated, falsifiable account of when KG-LLM hybrid grading helps. The more durable methodological lesson, though, may be the other half of the story: how readily an apparently strong robustness claim evaporated under a fairer control, and did so not on one dataset but on every one we checked it against.

References

[1] M. Mohler and R. Mihalcea, "Text-to-text semantic similarity for automatic short answer grading," in *Proc. 12th Conf. European Chapter ACL (EACL)*, Athens, Greece, 2009, pp. 567–575.

[2] M. Mohler, R. Bunescu, and R. Mihalcea, "Learning to grade short answer questions using semantic similarity measures and dependency graph alignments," in *Proc. 49th Ann. Meeting Assoc. Computational Linguistics (ACL)*, Portland, OR, USA, 2011, pp. 752–762.

[3] M. A. Sultan, C. Salazar, and T. Sumner, "Fast and easy short answer grading with high accuracy," in *Proc. 2016 Conf. North American Chapter ACL (NAACL-HLT)*, San Diego, CA, USA, 2016, pp. 1070–1075.

[4] M. Dzikovska et al., "SemEval-2013 Task 7: The Joint Student Response Analysis and 8th Recognizing Textual Entailment Challenge," in *Proc. 7th Int. Workshop Semantic Evaluation (SemEval)*, Atlanta, GA, USA, 2013, pp. 263–274.

[5] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. 2019 Conf. North American Chapter ACL (NAACL-HLT)*, Minneapolis, MN, USA, 2019, pp. 4171–4186.

[6] T. Sung, N. Nain, and H. K. Sharma, "Pre-trained language model-based automatic short answer grading for computer science education," *IEEE Access*, vol. 11, pp. 31 542–31 553, 2023.

[7] E. Emirtekin and Y. Özarslan, "Automated classification of student responses using Bloom's taxonomy with large language models," *Computers & Education: Artificial Intelligence*, vol. 8, 2025.

[8] B. S. Bloom, M. D. Engelhart, E. J. Furst, W. H. Hill, and D. R. Krathwohl, *Taxonomy of Educational Objectives: The Classification of Educational Goals. Handbook I: Cognitive Domain*. New York: David McKay, 1956.

[9] L. W. Anderson and D. R. Krathwohl (Eds.), *A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives*. New York: Longman, 2001.

[10] J. B. Biggs and K. F. Collis, *Evaluating the Quality of Learning: The SOLO Taxonomy*. New York: Academic Press, 1982.

[11] A. T. Corbett and J. R. Anderson, "Knowledge tracing: Modeling the acquisition of procedural knowledge," *User Modeling and User-Adapted Interaction*, vol. 4, no. 4, pp. 253–278, 1994.

[12] C. Pan, N. Li, M. Rusakov, and C. Faloutsos, "Prerequisite relation learning for concepts in MOOCs," in *Proc. 55th Ann. Meeting Assoc. Computational Linguistics (ACL)*, Vancouver, BC, Canada, 2017, pp. 1447–1456.

[13] N. Leacock and M. Chodorow, "C-rater: Automated scoring of short-answer questions," *Computers in Human Behavior*, vol. 19, no. 4, pp. 491–508, 2003.

[14] P. Bhandari et al., "Can ChatGPT replace human evaluators? An empirical study of automatic short answer grading," *arXiv preprint arXiv:2308.12505*, 2023.

[15] H. Gao et al., "Automated short answer grading using large language models: A zero-shot and few-shot exploration," in *Proc. 2024 IEEE Int. Conf. Advanced Learning Technologies (ICALT)*, 2024.
[16] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.